

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2018

Bachelor in Computer Application (BCA)
Course Title: MIS and E-Business

Group B

2. Explain C2C and B2B e-commerce models with example.
3. What is M-Commerce? Explain its benefits for organization, individual and society.
4. Define wireless communication. Differentiate between UMTS and LTE wireless data communication technologies.
5. Discuss how do enterprise application are used in organization.
6. What is digital signature? Explain the requirements of digital signature in secure transaction. How does digit signature work?
7. Explain cryptocurrency with its legal provisions in Nepal. How is it different from virtual currency?
8. What is Search Engine Optimization (SEO)? Explain the importance of SEO for the growth of a website.

Group C

9. Discuss the security requirements of Internet and E-commerce applications and how these requirements are fulfilled by various hardware and software systems? Explain in detail with examples.
10. "The use of electronic payment systems has significantly increased in Nepal in recent years". In this context, explain the trend of cashless transaction in Nepal along with the benefits and downsides of digital payment systems and cashless economy.
11. How can you evaluate whether the proposed e-commerce project is feasible or not. What can be the different funding techniques for a fresh graduate to setup a new online business? Explain

Tribhuvan University
Faculty of Humanities & Social Sciences
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2018

Bachelor in Computer Application (BCA)
Course Title: Computer Networking
Code No: CACS 303
Semester: V

Group B

2. Explain TCP/IP Reference model.
3. "Slotted ALOHA improves the performance of system over pure ALOHA". Justify.
4. Differentiate between Reliable & Unreliable connection with their application.
5. What is DNS? How it works?
6. What is cryptography? Explain the principles of cryptography.
7. What is TCP? Describe TCP segment header.
8. Write short note on (Any Two):
 - a) Channelization
 - b) Fragmentation
 - c) VPN

Group C

9. Classify and calculate the total number of Networks and Hosts per network in IPV4.
10. Explain the IPV4 Header format in detail.
11. What is switching? Describe the types of switching with their advantages and disadvantages.

Bachelor in Computer Application (BCA)
Course Title: Computer Graphics and Animation
Code No: CACS 305
Semester: V
2018

Group B

2. What do you mean by area filling? Explain boundary fill algorithm.
3. Define scaling transformation? Prove that two successive scaling are multiplicative.
- What is symmetric property of circle? Digitize a circle whose center is at origin and radius is 9.
- Define window and view port. Explain 2D viewing transformation pipeline.
6. Reflect a prism A(0,0,0), B(1,1,0), C(1,2,2) and (0,2,0) about yz-plane which has been rotated previously with +90 degree about y-axis.
7. What do you mean by visible surface detection? Explain Z buffer algorithm for visible surface detection.
8. What is animation? Explain the steps to create animation sequence.

Group C

9. Define refresh rate and persistence of monitor. Explain architecture of random scan display system with block diagram.
10. How can you draw line using Bresenham's line drawing algorithm? Trace it to rasterize the line with end points P (1, 4) and Q (7,8).
11. Explain Cohen Sutherland line clipping algorithm. Let ABCD be the rectangular window with A(20,20). B(90,20), C(90,70) and D(20,70). Use Cohen Sutherland algorithm to find the region codes and clip the lines with end point P(10,30) and Q(80,80)

Bachelor in Computer Application (BCA)
Course Title: Introduction to Management
Code No: CAMG 304
Semester: V
2018

Group B

11. Define management? What kinds of skills required for successful manager?
12. Describe the major characteristics of contingency theory. Why there is "no one best way" to manage
13. What is the planning? Explain the different level of planning.
14. Define organizational structure. What are the differences between centralization and decentralization?
15. Define communication and explain the various steps in the communication process.
16. What is leadership? Explain the different style of leadership.
17. What are the major technique of control? Explain.

Group C

18. Decision making is very important managerial function. With the statement; discuss the meaning and process of decision making. Also explain the types of decision making.
19. Quality improvement has no barriers and limitations. On the light of this statement, explain the concept of Total Quality Management and explain different tools of TQM.
20. Explain the concept of change in organization. Why do employees resist change in organization? And suggest various measures to overcome such resistivity to change.

Group B [6x5=30]

Attempts any SIX questions.

2. How properties like ubiquity, richness and information density make e-commerce better than traditional commerce? [1+4]
3. Differentiate B2C, B2B and C2C e-commerce with examples. [2+3]
4. What is the role of information superhighway in e-commerce? List the components of information superhighway. [2+3]
5. What is international information system? Describe the concepts of outsourcing and offshoring in the system. [1+4]
6. Define availability, authentication and authorization. How they can be enforced in e-commerce security? [3+2]
7. What is e-checking? Describe its working mechanism. [2+3]
8. Describe the website design criteria that one should show while designing an e-commerce website.

Group C

Attempts any TWO questions. [2x10=20]

9. What do you mean by Wireless Application Protocol (WAP)? How it works? Discuss the layered architecture of WAP. [2+2+6]
10. How important web contents are for e-commerce systems? Discuss their types. Justify, with examples, how web contents can be used to promote cross-selling, up-selling and promotions in e-commerce. [2+2+6]
11. Discuss in detail the security defends strategies that you can implement while securing e-commerce systems from security attacks and threats.

Group B [6x5=30]

Attempts any SIX questions

2. Define protocols. Explain WWW and HTTP protocol. [1+2+2]
3. Define transmission impairment. Explain the causes of impairments. [1+4]
4. Define HDLC. Explain the HDLC frame formats. [1+4]
5. Define IP Address. Specify IPv4 address classes with their address ranges. [1+4]
6. Define Subnetting. Suppose you are given network address: 192.168.10.0 and subnet mask: 255.255.255.240 then calculate total number of subnets and numbers of hosts per subnet. [1+2+2]
7. Draw a User Datagram format. Explain UPD operations. [2+3]

8. Write short notes on (Any Two): [2.5+2.5]

a) DNS

b) Public Key Cryptography

c) VPN

Group C

Attempt any TWO questions. [2x10=20]

9. Critically analyze the OSI reference model. [10]
10. Explain the random-access protocols under the multiple access taxonomy. [10]
11. Explain the IPv4 Header format in detail. [10]

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2020
Bachelor in Computer Application (BCA)
Course Title: MIS and E-Business

Group B [6x5=30]

Attempts any SIX questions.

2. How properties like ubiquity, richness and information density make e-commerce better than traditional commerce?
3. Differentiate B2C, B2B and C2C e-commerce with examples.
4. What is the role of information superhighway in e-commerce? List the components of information superhighway. [2+3]
5. What is international information system? Describe the concepts of outsourcing and offshoring in the system. [1+4]
6. Define availability, authentication and authorization. How they can be enforced in e-commerce security? [3+2]
7. What is e-checking? Describe its working mechanism. [2+3]
8. Describe the website design criteria that one should show while designing an e-commerce website.

Group C

Attempts any TWO questions. [2x10=20]

9. What do you mean by Wireless Application Protocol (WAP)? How it works? Discuss the layered architecture of WAP. [2+2+6]
10. How important web contents are for e-commerce systems? Discuss their types. Justify, with examples, how web contents can be used to promote cross-selling, up-selling and promotions in e-commerce. [2+2+6]
11. Discuss in detail the security defends strategies that you can implement while securing e-commerce systems from security attacks and threats.

Tribhuvan University
2020
Bachelor in computer Application(BCA)
Course Title: Introduction to management
Semester 5th

Group B

2. What is management? Explain the functions of management.
3. Define scientific management. Explain the major contributions made by F.W. Taylor.

4. Describe the formal and informal group with suitable example.
5. What is planning? Discuss the steps involved in planning process.
6. What are the barriers of effective communication in any organizations? Explain.
7. Why is motivation important to achieve organizational goals?
8. Define delegation of authority. Explain major features of delegation of authority.

Group C

Attempts any TWO questions.

9. Planning is head and controlling is tail. Justify.
10. Define leadership. Explain different style of leadership with suitable example.
11. What are managerial skills? What skills are essential for a successful manager.

Bachelor in Computer Applications

Course Title: DotNet Technology

Code No: CAC8302

Semester: V

2021

Group B

2. What do you mean C# is type safety programming language? Explain any four applied technologies in Dot Net framework.
3. List any five contextual keywords in C#? Write a C# program to initialize and display jagged array elements with sum of each row. [1+4]
4. What is string interpolation? How passing argument by value is differing from passing argument by reference? Explain with program. [1 4]
5. What are uses of base keyword in C#? Write a CW program to illustrate the concept of abstract class and abstract method. [1-2--+2]
6. What is virtual method? Write a C# program to illustrate multi-level inheritance with virtual method. [1+4]
7. What is difference between indexer and properties? Explain the concept of generic class in C# with suitable program. [2+3]
8. Write a program to create form for calculating simple interest in one ASP.NET page and display the simple interest in another page of ASP.NET. [5]

Group C

9. a) What is SystemException? Write a C# program that will read balance and withdraw amount from keyboard and display the remaining balance on screen if balance is greater than withdraw amount otherwise throw an ApplicationException with appropriate message. [1+4]
b) List any five LINQ standard operator? Write a C# program to compute aggregate salary of 5 employee and then display employee record in descending order with respect to employee salary using LINQ. | 1+4]
10. What are different type of delegate in C#? Write a C# program to create a class Time which represents time. The class should have three fields for hours, minutes and seconds. It should have constructor to initialize hours, minutes and seconds and method displayTime() to print current time. Overload following operators:
a) + (add two time objects on 24 hours clock)
b) < (compare two time objects)

11. What is difference between executeScalar() and executeReader() method ?
Write a C# program to connect database Bank and insert 5 customer record (Account_no, Name, Address, Balance) in Customer table and Display the customer record whose balance is greater than 5000.

Batch: 2019
Bachelor in Computer Applications
Course Title: Introduction to Management
Code No: CAMG304
Semester: V

Group B

2. Define management. Describe the functions of management.
3. What is planning? Discuss the process of planning.
4. What are the condition of decision making? Explain.
5. Discuss the steps of controlling.
6. What are the main leadership styles? Explain.
7. Describe the major important techniques that can be used for motivating employees.
8. Define communication? Explain the types communication.

Group C

9. Explain contingency theory of management. What are its contributions and limitations?
10. "Organizing structuring is the process of designing job, grouping jobs into manageable units, and establishing patterns of authority among jobs and groups of jobs". In line of this statement state the steps involved in the process of structuring an organization. Also explain the principle of organizing.
11. Change can be introduced successfully only when there are conducive factors in existence in the system. In the highlight of the statement discuss the paradigm shifts and areas of organizational change. Also explain the factor overcoming resistant to organizational change.

Bachelor in Computer Applications
Course Title: MIS and e-Business
Code No: CACS301
Semester: V
2021

Group B

2. What has Location-based Commerce done to help online business? Explain its main idea, components, and the applications. [5]
3. Define wireless communication. What are the features of Wi-Fi that have made it so popular in the present? [1+4]
4. How do systems for managing relationships with customer's help businesses get closer to their customers?
5. Differentiate between virus, worm and Trojan horse. [5]
6. What is encryption and decryption? Explain the symmetric key encryption and asymmetric key encryption in detail. [1+4]
7. What is virtual currency? Explain the different types of virtual currencies popular in the market.
8. What are the different ways an e-commerce website can be promoted so that it has a strong presence in the market? Explain. [5]

Group C

9. Define e-Commerce. Explain its significance in today's business world. Explain different types of e-commerce models with examples. [2+3+5]
10. "In the digital world we live in now, information systems like Transaction Processing System, Management Information System, Decision Support System, and Executive Support System are a must for any organization." Explain why the statement is true, including what these information systems are and how they help the organization. [10]
11. As more organizations use IT systems, ransomware attacks have become a big problem, especially during and after the Covid 19 pandemic. In this scenario, describe ransomware in detail, including the effects of attacks, different types of ransomware, ways to protect against them, and what to do when they happen. [10]

Faculty of Humanities & Social Sciences
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2021

Bachelor in Computer Applications
Course Title: Computer Networking

Code No: CACS303

Semester: V

Group B

2. What is digital signature and how it works?
3. Why do we need DNS when we can directly use an IP address? Explain the process to resolve the domain www.tribhuvan-university.edu.np ? [2+3]
4. What do you mean by IP datagram fragmentation?
5. Explain the mechanism of bit and byte stuffing with suitable example.
6. Explain the causes of transmission impairment in data communication.
7. What is flow control? Explain Go-Back-N ARQ with suitable diagram. [1+4]
8. Explain in detail about the structure of IPv6 address. Differentiate between IPv4 and IPv6 address. [3+2]

Group C

9. Explain how slotted Aloha improves the performance of system over pure Aloha. A bit stream of 100101010 is transmitted using CRC method. The generator polynomial is $x^3 + 1$. Show the actual bit string transmitted and if fourth bit from LSB is altered, show the error detection in receiver side. [2+8]
10. Why do we need layered protocol architecture? Discuss each layer of TCP/IP architecture along with function of each layer. Write the protocols used in each layer of TCP/IP model. [2+4+4]
11. Why do we need subnetting? What is the new subnet mask? Write subnet ID and broadcast address of each subnet if you divide a class C network (192.6.3.0-192.6.3.255) into 4 different subnets?

Course Title: Computer Graphics and Animation (3 Cr.)

Course Code: CACS305

Year/Semester: III/V

Class Load: 6 Hrs. / Week (Theory: 3Hrs. Tutorial: 1 Hrs., Practical: 2 Hrs.)

Course Description

This course is designed to extend students' knowledge and practice in Graphics hardware, software, and applications. It also provides the knowledge of data structures for graphics, graphics languages, and models for 2D and 3D objects, clipping, hidden surface elimination, depth buffer, raster graphics, shading, and rendering.

Course objectives

Upon completion of this course, students should be able to 1. Explain basic principle of computer graphics. 2. Develop 2D and 3D computer graphics applications. 3. Specify lighting and object's materials in computer graphics programming.

Course Contents

Unit 1: Introduction

6

- 1.1 Advantage of Computer Graphics and Areas of Applications
- 1.2 Hardware and Software for Computer Graphics. (Hard Copy, Display Technologies),
- 1.3 Random Scan Display System, Video Controller, Random Scan Display Processor
- 1.4 Raster Graphics
- 1.5 Scan Conversion Algorithms (Line, Circle, Ellipse)
- 1.6 Area Filling (Rectangle, Ellipse), Clipping (Lines, Circle, Ellipse), Clipping Polygons

Unit 2: Two dimensional and three dimensional transformations

7

- 2.1 2-Dimensional transformation
- 2.2 2-D Translation, Rotation, Scaling,
- 2.3 Homogeneous Coordinates, Reflection, Shear transform
- 2.4 3-dimensional transformation,
- 2.5 3-D Translation, Rotation Scaling, Reflection, Shear.

Unit 3: Clipping

7

- 3.1 Window to view port transformation
- 3.2 Clipping, line clipping,
- 3.3 Cohen –Sutherland line clipping
- 3.4 Polygon clipping
- 3.5 Sutherland and Gary Hodgman polygon clipping algorithm



Unit 4: Visible Surface Determination and computer graphics algorithm

15

- 4.1 Image space and object space techniques
- 4.2 Hidden Surface removal—Depth comparison
- 4.3 Z-Buffer Algorithm
- 4.4 Back-Face Removal
- 4.5 The Painter's Algorithm
- 4.6 Scan-Line Algorithm
- 4.7 Light and Color and different color models (RGB, CMY, YIQ)

Unit 5: Animation and virtual reality

10

- 5.1 Basic Principles of Animation and Types of Animation
- 5.2 Introduction to the flash interface
- 5.3 Setting stage dimensions, working with panels, panel layouts
- 5.4 Layers & Views
- 5.5 Shaping Objects – Overview of shapes, Drawing & Modifying Shapes
- 5.6 Bitmap Images & Sounds
- 5.7 Animation -Principles , Frame by frame animation, tweening, masks
- 5.8 Introduction to virtual reality

Laboratory Work

Laboratory work should be done covering all the topics listed above and a small project work should be carried out using the concept learnt in this course using Open GL.

Reference Books:

1. Foley, J. D., A. V. Dam, S. K. Feiner, J. F. Hughes, Computer Graphics Principle and Practices, Addison Wesley Longman, Singapore Pvt. Ltd.,
2. Hearn Donald, M. P. Baker, Computer Graphics, 2E, Prentice Hall of India Private Limited, New Delhi
3. Robert R & Snow D Flash CS4 Professional Bible, Wiley Publishing



Course Title: Computer Networking (3 Cr.)

Course Code: CACS303

Year/Semester: III/V

Class Load: 5 Hrs. / Week (Theory: 3Hrs. Practical: 2 Hrs.)

Course Description

This course offers detailed concept and structure of networking standards and principles. It includes introduction, functioning and significance of Physical Layer, Data Link Layer, Network Layer, Transport Layer, Application layer and some security mechanisms. It does not entirely focus on theoretical concept but also strongly focuses on practical skill based learning.

Course objectives

The general objectives of this course are to provide theoretical as well as practical knowledge of computer networking to make students capable of implementing, managing and troubleshooting the issues of computer network in their personal as well professional life.

Course Contents

Unit 1: Introduction

6 Hrs.

- 1.1 Network as an infrastructure for data communication
- 1.2 Applications of Computer network
- 1.3 Network Architecture
- 1.4 Types of computer Networks
- 1.5 Protocols and Standards
- 1.6 The OSI Reference Model
- 1.7 The TCP/IP Protocol Suite
- 1.8 Comparison between OSI and TCP/IP Reference model
- 1.9 Critiques of OSI and TCP/IP Reference model

Unit 2: The Physical Layer

6 Hrs.

- 2.1 Functions of Physical Layer
- 2.2 Data and Signals: Analog and Digital signals, Transmission Impairment, Data Rate Limits, Performance
- 2.3 Data Transmission Media: Guided Media, Unguided Media and Satellites
- 2.4 Bandwidth Utilization: Multiplexing and Spreading
- 2.5 Switching: Circuit switching, Message switching & Packet switching
- 2.6 Telephone, Mobile and Cable network for data Communication

Unit 3: The Data Link Layer

8 Hrs.

- 3.1 Functions of Data Link Layer
- 3.2 Data Link Control: Framing, Flow and Error Control
- 3.3 Error Detection and Correction
- 3.4 High-Level Data Link Control(HDLC) & Point – to – Point protocol(PPP)
- 3.5 Channel Allocation Problem



- 3.6 Multiple Access: Random Access (ALOHA, CSMA, CSMA/CD, CSMA/CA), Controlled Access (Reservation, Polling, Token Passing), Channelization (FDMA, TDMA, CDMA)
- 3.7 Wired LAN: Ethernet Standards and FDDI
- 3.8 Wireless LAN: IEEE 802.11x and Bluetooth Standards
- 3.9 Token Bus, Token Ring and Virtual LAN

Unit 4: The Network Layer

8 Hrs.

- 4.1 Functions of Network Layer
- 4.2 Virtual circuits and Datagram Subnets
- 4.3 IPv4 Addresses: Address Space, Notations, Classful addressing, Classless addressing, Subnetting and Network Address Translation (NAT)
- 4.4 IPv4 Datagram format and fragmentation
- 4.5 IPv6 Address Structure and advantages over IPv4
- 4.6 Routing Algorithms: Distance Vector Routing, Link State Routing
- 4.7 Internet Control Protocols: ARP, RARP, ICMP
- 4.8 Routing protocols: OSPF, BGP, Unicast, Multicast and Broadcast

Unit 5: The Transport Layer

7 Hrs.

- 5.1 Functions of Transport Layer
- 5.2 Elements of Transport Protocols: Addressing, Establishing and Releasing Connection, Flow Control & Buffering, Error Control, Multiplexing & Demultiplexing, Crash Recovery
- 5.3 User Datagram Protocol (UDP): User Datagram, UDP Operations, Uses of UDP, RPC
- 5.4 Principles of Reliable Data Transfer: Building a Reliable Data Transfer Protocol, Pipelined Reliable Data Transfer Protocol, Go-Back-N (GBN), Selective Repeat (SR)
- 5.5 Transmission Control Protocol (TCP): TCP Services, TCP Features, TCP Segment Header
- 5.6 Principle of Congestion Control

Unit 6: The Application Layer

5 Hrs.

- 6.1 Functions of Application layer
- 6.2 Application Layer Protocols: DNS, DHCP, WWW, HTTP, HTTPS, TELNET, FTP, SMTP, POP, IMAP
- 6.3 Concept of traffic analyzer: MRTG, PRTG, SNMP, Packet tracer, Wireshark.

Unit 7: Network Security

5 Hrs.

- 7.1 A Model for Network Security
- 7.2 Principles of cryptography: Symmetric Key and Public Key
- 7.3 Public Key Algorithm – RSA
- 7.4 Digital Signature Algorithm
- 7.5 Communication Security: IPsec, VPN, Firewalls, Wireless Security.



Practical

1. Prepare hardware and software specification for basic computer system.
2. Determine the appropriate placement of networking devices on a network.
3. Identify networking cable standards. Create and test cross – over and straight cables.
4. Configure the IP address of the computer.
5. Create a basic network and share file and folders.
6. Install and configure windows server: Active Directory, User and Group Policy Management.
7. Set the file access permissions and quota in windows server.
8. Configure basic DNS and DHCP services in windows server.
9. Install Linux based OS and practice on basic Linux and networking commands.
10. Configure IP address and subnet in Linux Machine.
11. Install packet tracer and identify the features of packet tracer.
12. Implement the LAN topologies.
13. Demonstrate the use of VLAN.
14. Implement the both static and dynamic router configurations.
15. Install and configure DNS, DHCP, FTP and Web Servers in Linux machine.
16. Capture some packets and analyze the header using Wireshark.
17. Implement the firewall.

Teaching Methods

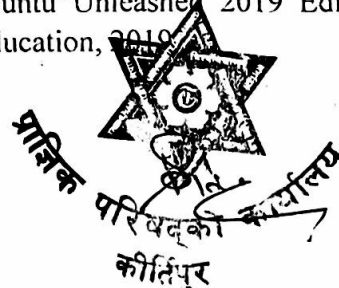
The teaching faculties are expected to create environment where students can update and upgrade themselves with the current scenario of computing and information technology with the help of topics listed in the syllabus. The general teaching pedagogy that can be followed by teaching faculties for this course includes class lectures, laboratory activity, group discussions, case studies, guest lectures, research work, project work, assignments (Theoretical and Practical), and written and verbal examinations.

Evaluation

Examination Scheme				
Internal Assessment		External Assessment		Total
Theory	Practical	Theory	Practical	
20	20 (3 Hrs.)	60 (3 Hrs.)	-	

Reference Book

1. Andrew S. Tanenbaum, David J. Wetherall, "Computer Networks, 5/e", Prentice Hall, 2011.
2. Behrouz A. Forouzan, "Data Communications and networking" Tata McGraw-Hill.
3. Kurose, Ross, "Computer Networking: A Top-Down Approach", Pearson Education Limited, 2017.
4. Larry L. Peterson and Bruce S. Davie, "Computer Network: A System Approach", Morgan Kaufmann, 5/e, 2012.
5. Matthew Helmke, Andrew Hudson, Paul Hudson "Ubuntu Unleashed 2019 Edition_ Covering 18.04, 18.10, 19.04", 13/e, SAMS _ Pearson Education, 2019.





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OFFICE OF THE DEAN

2018

Bachelor in Computer Application (BCA)
Course Title: Computer Graphics and Animation
Code No: CACS 305
Semester: V
Symbol No:

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Center:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt All Questions:

[10×1=10]

1. Circle (O) the correct answer in the following questions:

- a) Random scan system is used for
 - i. Color drawing application
 - ii. Line drawing application
 - iii. Pixel drawing application
 - iv. All of the above
- b) Which of the following is used to store picture definition in random display system?
 - i. Display Controller
 - ii. Vertical retrace
 - iii. Refresh display file
 - iv. Frame buffer
- c) Which of the following algorithm is used to fill the interior of polygon?
 - i. Parallel curve algorithm
 - ii. Flood fill algorithm
 - iii. Scan line fill algorithm
 - iv. Boundary fill algorithm
- d) Which of the following equation is used in 2D scaling of point $P(x,y)$ to $P'(x',y')$
 - i. $x'=x*s_x$ and $y'=y*s_y$
 - ii. $x'=y*s_x$ and $y'=x*s_y$
 - iii. $x'=x+s_x$ and $y'=y+s_y$
 - iv. $x'=x+s_y$ and $y'=y+s_x$
- e) What is the value of initial decision parameter in the midpoint circle drawing method?
 - i. $\frac{4}{5}-r$
 - ii. $\frac{3}{5}-r$
 - iii. $\frac{5}{3}-r$
 - iv. $\frac{5}{4}-r$
- f) Which of the following hidden surface removal algorithm does not employ image space approach?
 - i. Depth Buffer method
 - ii. A buffer method
 - iii. Back face detection method
 - iv. Scan line method
- g) In Cohen Sutherland line clipping, 4 bit region code of bottom left region of window is:
 - i. 1100
 - ii. 0101
 - iii. 101
 - iv. 1001
- h) Which of the following is the space in which application model is defined?
 - i. World coordinate system
 - ii. Screen coordinate system
 - iii. Vector coordinates system
 - iv. Interface window
- i) The intersection of primary colors in CMYK color model generates:
 - i. Green color
 - ii. Blue color
 - iii. Dark red color
 - iv. Black color
- j) Which of the following allows you to insert text within your flash stage?
 - i. Text box
 - ii. Text tool
 - iii. Pen tool
 - iv. Oval tool



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Code No: CACS 305
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any six questions:

[6×5=30]

2. What do you mean by area filling? Explain boundary fill algorithm.
3. Define scaling transformation? Prove that two successive scaling are multiplicative.
4. What is symmetric property of circle? Digitize a circle whose center is at origin and radius is 9.
5. Define window and view port. Explain 2D viewing transformation pipeline.
6. Reflect a prism A(0,0,0), B(1,1,0), C(1,2,2) and (0,2,0) about yz-plane which has been rotated previously with +90 degree about y-axis.
7. What do you mean by visible surface detection? Explain Z buffer algorithm for visible surface detection.
8. What is animation? Explain the steps to create animation sequence.

Group C

Attempt any two questions:

[2×10=20]

9. Define refresh rate and persistence of monitor. Explain architecture of random scan display system with block diagram.
10. How can you draw line using Bresenham's line drawing algorithm? Trace it to rasterize the line with end points P (1, 4) and Q (7, 8).
11. Explain Cohen Sutherland line clipping algorithm. Let ABCD be the rectangular window with A(20,20), B(90,20), C(90,70) and D(20,70). Use Cohen Sutherland algorithm to find the region codes and clip the lines with end point P(10,30) and Q(80,80)



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2021

Bachelor in Computer Applications
Course Title: Computer Graphics and Animation
Code No: CACS305
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 hours
Batch: 2019

Centre:

Symbol No:

Candidates are required to answer the questions in their own words as far as possible.

Group A

Attempt all the questions.

[10×1 = 10]

1. Circle (O) the correct answer.

- i) Random scan systems generally use ____ to display an image
a) Memory b) Frame buffer c) Processor d) Refresh Display File
- ii) Which of the following is used for the area of the computer captured by an application?
a) Viewport b) Display c) Window d) Buffer
- iii) Which color is produced by the intersection of RGB
a) Blue b) White c) Green d) Red
- iv) In the back-face detection algorithm, if V is along positive Z direction than polygon is Back Face if.....
a) $C \geq 0$ b) $C \leq 0$ c) $C \geq 1$ d) $C \leq 1$
- v) How many steps are required to scale an object about a fixed point?
a) 1 b) 2 c) 4 d) 3
- vi) Which types of lines are ignored in the scan line algorithm to fill regions?
a) Horizontal b) Vertical c) Left Inclined d) Right Inclined
- vii) Reflection of a point about X-axis followed by a counter clockwise rotation of 90 degree is equivalent to reflection about the line?
a) $x = -y$ b) $y = -x$ c) $x = y$ d) $x + y = 1$
- viii) What is the 4-bit code of the top left-region among the nine regions divided using the Cohen-Sutherland algorithm?
a) 0001 b) 1001 c) 0110 d) 0000
- ix) Which of the following equation is used in 2D scaling to scale a point(x,y) to the new point (x',y')?
a) $x' = x \times s_x$ and $y' = y \times s_y$ b) $x' = x + s_x$ and $y' = y + s_y$
c) $x' = x \times s_y$ and $y' = y \times s_x$ d) $x' = x + t_x$ and $y' = y + t_y$
- x) What part of flash menu bar allows you to hide/unhide panels?
a) View b) Edit c) Window d) Layer Panel



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2021

Bachelor in Computer Applications
Course Title: Computer Graphics and Animation
Code No: CACS305
Semester: V

Full Marks: 60
Pass Marks: 24
Time: 3 hours
Batch: 2019

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Define horizontal and vertical retrace of electron beam. Explain architecture of random scan display system. [2+3]
3. How does decision parameter can be used to draw circle? Calculate the points to draw a circle having radius 5 and center has (10, 5) using mid-point circle algorithm. [1+4]
4. What do you mean by window to viewport transformation? Explain 2D viewing pipeline. [2+3]
5. Generate the points between the end points of a line viz. (2, 2) and (9,6) by using DDA line drawing algorithm.
6. Define visible surface detection. Outline the Z-buffer algorithm along with its advantages and disadvantages. [1+4]
7. What do you mean by Virtual Reality? Explain the types of Virtual Reality. [2+3]
8. Define reflection transformation and derive the 2D reflection matrix along x-axis and y-axis in homogeneous coordinate. [1+4]

Group C

Attempt any TWO questions.

[2×10 = 20]

9. What is the need of homogeneous coordinate system in geometric transformation system? Find the new co-ordinate of rectangle ABCD whose center is at (4, 2) is reduced to half of its size and center will remain same. The co-ordinate of ABCD are A(0, 0), B(0, 4), C(8, 4) and D(8, 0). [2+8]
10. Explain Sutherland Hodgeman Polygon Clipping algorithm and trace it to clip a polygon with end points P(10, 30), Q(25, 35), R(35,50), S(60,30), T(45, 28), U(40,15) against a window whose lower left corner is at (15, 25) and upper right corner is at (50, 40). [4+6]
11. What do you understand by object space and image space method? Explain Flood Filled algorithm with its applicability and concept. [4+6]



Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
2024

Bachelor in Computer Applications

Course Title: Computer Graphics and Animation

Code No: CACS 305

Semester: V

Full Marks: 60

Pass Marks: 24

Time: 3 hours

Candidates are required to answer the questions in their own words as far as possible.

Group B

Attempt any SIX questions.

[6×5 = 30]

2. Differentiate between beam penetration method and shadow mask method. How does raster scan display system work? [3+2]
3. How DDA works? Digitize the line segment with the endpoints (15, 25) and (21, 35) using DDA line drawing algorithm. [2+3]
4. What is a viewport? Consider a window with lower left corner at (2, 2) and upper right corner (5, 10) and a viewport with left lower corner at (5, 5) and upper right corner at (8, 8). What will be the value of the point in the viewport after the window to viewport transformation if the point is (4, 4) in the window? [1+4]
5. Explain how scan line algorithm can be used for hidden surface removal. [5]
6. Derive the transformation matrices for 3D rotation and reflections. [2.5+2.5]
7. What is animation? Explain about applications of Virtual Reality. [1+4]
8. Write short notes on (any two): [2.5+2.5]
 - a) RGB color model
 - b) 2D shear transformation
 - c) Computer Graphics Vs. Image Processing

Group C

Attempt any TWO questions.

[2×10 = 20]

9. Differentiate between boundary fill algorithm and flood fill algorithm in detail. Find the composite transformation matrix for anti-clockwise rotation of 60° about a point (2, 3). Use it to rotate a triangle ABC with vertices A (4, 3), (5, 5) and (8, 9). [5+3+2]
10. What is visible surface detection? How object space method can be used for visible surface detection? Explain depth-sorting method for visible surface detection in detail. [2+2+6]
11. How region codes are used in Cohen-Sutherland algorithm to clip the line segments? Find the clipping coordinates for lines PQ and RS with P(20, 5), Q(40, 30) and R(20, 20), S(80, 60) against the window with left upper corner(10, 50) and right lower corner(50, 10). [4+6]

Tribhuvan University
Institute of Science and Technology
2020

Bachelor Level / Fifth Semester / Science

Computer Science and Information Technology(CACS305)

((TU BCA) Computer Graphics and Animation)

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Full marks: 60

Pass marks: 24

Time: 3 hours

Group B

Attempts any SIX questions.

[6x5=30]

2. What is computer graphics? Explain different application areas of computer graphics.
3. How can you draw a circle using mid-point circle algorithm? Explain with the algorithm.
4. Explain 3D basic geometric transformation with an example.
5. What is polygon clipping? Explain Sutherland Hodgman algorithm for polygon clipping.
6. Given a triangle with vertices A(2,3), B(5,5), C(4,3) by rotating 90 degree about the origin and then translating two unit in each direction. Use homogenous transformation matrix to find the new vertices of the triangle.
7. Explain the scan line algorithm for visible surface detection.
8. Explain the architecture of VR system with necessary components.

Group C

Attempts any TWO questions.

[2x10=20]

9. Explain the working details of DDA line drawing algorithm? Digitize the line with endpoints (2,2) and (10,5) using Bresenham's line drawing algorithm. [5+5=10]
10. Write the different between object space method and image space method. Explain Z buffer algorithm for visible surface detection. [5+5=10]
11. Derive the formula for windows to viewport transformation. Given a window bordered by (0,0) at the lower left and (4,4) at the upper right. Similarly, a viewport bordered by (0,0) at the lower left and (2,2) at the upper right. If a window at position (2,4) is mapped into the viewport. What will be the position of viewport to maintain same relative placement as in window? [5+5=10]

Course Title: Introduction to Management (3 cr.)

Course code: CAMG 304

Year/Semester: III /V

Class load: 3 Hrs./Week (Theory: 3Hrs)

Course Objectives

This course aims to impart the basic management knowledge, and skills to the students so as to enhance their managerial capabilities and enable them to apply in the practical field.

Course Description

This course contains Introduction to Management, perspectives in management thought, emerging issues and challenges in management, management functions like planning, leading, controlling, organizational change and development, communication, emerging issues in quality management, technology and management.

Course Contents

Unit 1: Introduction

LH 4

Management: concepts, meaning and functions. Types of managers. Managerial roles and skills. Organization and management. Changing perspectives of organization.

Unit 2: Perspectives in Management

LH 7

Classical Perspective: scientific management, administrative management and bureaucracy. Behavioral Perspective: Hawthorne studies, human relations movement, and emergence of organizational behavior. Quantitative Perspective: management science and operations management. Integrating perspectives: systems and contingency perspectives. Emerging management issues and challenges.

Unit 3: Planning and decision making

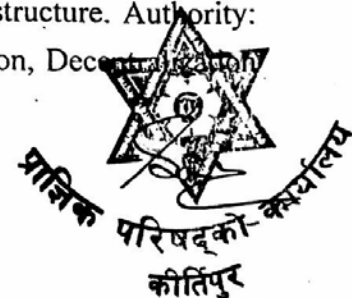
LH 7

Concept of planning, Levels of Planning: Strategic, Tactical and operational. Steps in Planning. Tools for planning. Decision Making: meaning, types and process. Decision making conditions – certainty, risk and uncertainty.

Unit 4: Organizing

LH 9

Concept of organizing, process and principles of organizing. Organization Architecture: vertical differentiation – tall versus flat hierarchies, horizontal differentiation – functional structure, multidivisional structure, geographic structure, and matrix structure. Authority: line authority and staff authority. Delegation of authority. Centralization, Decentralization



and Devolution: meaning, reasons, advantages and disadvantages. Staffing: concept and importance.

Unit 5: Leading and communication

LH 7

Concept and qualities of leadership. Transformational and transactional leadership, Leadership Styles: autocratic, democratic, and participative. Concept of managerial ethics. Motivation: concept, importance, and techniques. Communication: meaning, process, and networks. Types of communication, Barriers to effective communication.

Unit 6: Controlling and total quality management

LH 5

Concept, purpose, Process and types of controls. Essentials of effective control systems. Control tools and techniques. Quality: Concept and importance. Total Quality Management: concept, components, principles, tools and techniques. Emerging issues in quality management.

Unit 7: Organizational Change and Development

LH 5

Concept and nature, forces, paradigm shifts and areas (structure, technology, business process and behaviors) of organizational change. Resistance to change. Overcoming resistance to change. Concept of Organizational Development

Unit 8: Technology, Organization and Management

LH 4

Concept of technology, approaches to technology and organization, social networking, use of technology in people management,

References

Charles W.L. Hill and Steven L. McShane, *Principles of Management*, Tata Mc-Graw-Hill Company, New Delhi.

Griffin, Ricky W., *Management*. AITBS Publishers and Distributors, New Delhi.

Hitt, M.A., J.S. Black and Porter, L.W., *Management*, Pearson Education, New Delhi

Laurie J. M. *Management and organizational Behaviour*, Pearson, New Delhi

Evaluation:

Internal Assessment: 40 marks

External Assessment: 60 marks

Total: 100 marks

Teaching methods: The major teaching methods include the case analysis, project work, term paper, assignments, and quiz. The instructor decides the learning strategies based on the nature of session/class.



Course Title: MIS and E-Business

Course Code: CACS301

Year/Semester: III/V

Class Load: 5Hrs. /Week (Theory: 3Hrs, Practical 2Hrs.)

Course Description

This course provides the fundamental knowledge of Management Information System and E-Business and its associated infrastructure, security and protection issues, payment system and ultimately live implementation of any web application in Web server.

Objectives: The general objectives of this course is to know fundamental knowledge of MIS/E-Business and implement the web application in Web Server.

Unit 1:

Introduction to E-Commerce: Definitions and Concepts; Defining E-Business; Pure Versus Partial EC; EC Framework, Classification of EC; Benefits of E-Commerce; Electronic Markets; Role of Internet and Web in E-Commerce; The Limitations and Barriers of EC; Social Networks and Social Network Services; **M-Commerce:** Concept, Scope, Attributes, Benefits; Location-based I-commerce, I-Commerce Infrastructure, Location-Based Services and Applications. 7LH

Unit 2:

The Network Infrastructure for E-Commerce: Introduction to Information Superhighway (I-Way), Components of the I-Way, Internet as a network infrastructure. **Wireless Application Protocol:** Wireless Application Protocol (WAP); Architecture of WAP; Working of WAP; Wireless Technologies: ADSL, WiMAX, WLAN, WMAN Wi-Fi, UMTS (3G), LTE (4G), (5G NR). Security Issues related to Wireless Communications. 8LH

Unit 3: Introduction to Management Information System: Data, information, computer based information system (CBIS), Information System Resources, Management information system, Transaction processing (TPS) system, decision support system (DSS), and executive information system (EIS), *SCM, CRMS and International Systems:* Introduction, Supply Chain Management Systems, Customer Relationships Management Systems, enterprise systems and Challenges of Enterprise Systems Implementations- Managing the implementation, International Information Systems-Outsourcing and off-shoring. 8LH

Unit 4: E-Commerce Security and Fraud Issues and Protections: Basic EC Security Terminology, The Threats, Attacks, and Attackers, **EC Security Requirements:** Confidentiality, Integrity, and Availability, Authentication, Authorization and Nonrepudiation; Technical Malware attack: Viruses, Worms, and Trojan Horses, Heartbleed, Distributed Denial of Service, Cryptoblocker, Page hijacking, Botnets, Malvertising, ransomware, sniffing; Non-Technical Malware attack: Social Phishing, Pharming, Identity



Theft and Identify Fraud, Spam Attacks; EC defense Strategy: access control(Authorization and Authentication, Biometric Systems), encryption and PKI (Symmetric Key Encryption, Asymmetric Key Encryption, Certificate Authority(CA), Secure Socket Layer (SSL). Securing e-commerce networks: Firewalls, Virtual Private Networks, Intrusion Detection Systems (IDS), intrusion prevention System (IPS). **10LH**

Unit 5: E-payment systems:

Online payment cards (credit cards, charge cards, debit cards, smart cards), processing cards in online, credit card payment procedure, e-micropayments, e-checking and its processing in online. Automated clearing house (ACH) network, mobile payments (Digital wallet), mobile payment participants and issues, international payments, emerging EC payment systems and issues: crypto currency, virtual currency. A case study of emerging trends in online payment system in Nepal. **6 LH**

Unit 6: Launching a Successful EC Website:

Planning Online Businesses (Business Plan, The cost-benefit, risk elements of a business case, funding a New Online Business, EC Model selection), The process of building a website, basic hierarchical website structure, website hosting and obtaining a domain name (cPanel management, upload EC site to the web Server), web content creation and management: (Web content, Categories and Types of Content, Content Management and Maintenance, Catalog Content and its Management), Website design criteria, Site Map and Navigation, Web page layout grid, Colors and Graphics, Website Usability, Performance, Website Promotion, implementation of Payments system, Website Promotion, Search Engine Optimization (SEO). **10LH**

Laboratory Works:

32LH

Laboratory works should be done covering all the topics listed above and a project work should be carried out by individually implementing a fully functioning e-commerce web application along with payment system.

Text Book

Efraim Turban, D. K. (2018). *Electronic Commerce: A Managerial and Social Networks Perspective*. New York: Springer

References Book

Chaffey, D. (2009). *E-Business and E-Commerce Management: Strategy, Implementation and Practice (4th Edition)*. Harlow: Prentice Hall.

Kalakota, A. B. (1996). *Frontiers of Electronic Commerce*. Pearson.

Kenneth C. Laudon, C. G. (2014). *E-commerce: business.technology.society*. New York: Pearson Education Limited.

Kenneth C. Laudon, J. P. (2014). *Management Information System: MANAGING THE DIGITAL FIRM*. New York: Pearson Education Limited.

